COM 126.

Pc Upgrade and Maintenance.

Introduction: In today's technologically driven World, People depends almost fully on their Computer to perform even the Simplest task. Businesses maintain maximum performance through the Use of the most Up-to-date technologies; Schools Keeps better managment of Students Record's through the use of Computers and even any person Can benefit from these Machines.

However, all these efficiencies will be lost If Computers are not maintained.

Maintenance involves operation and functional Checks, Servicing, repairing/replacing of necessary Component and supporting Utilities of Pc so that it could be kept in a good working Condition

Importance of Maintenance:

Some Importance of Computer mentainance will Include but not limited to the following;

1. It extend the Usefull life of a Computer.
2. It Improves productivity.
3. It ensures steady operation of the Computer
4. It minimizes Lost.
5. Higher return on output.

TYPES OF MENTAINANCE.

1. Predictive Mentainance
2. Preventive Mentainance
3. Corrective Mentainace
4. Evolutionary Mentainance

1. Predictive Mentainance:

More recently, advance in Sensing and Computing technology have given rise to predictive Mentainan
The mentainance strategy Uses Sensors to monitor key parameters withing a machine and uses this data in Conjuction with analysed historical trend to Continuosly evaluate th system health and predict a breakdown befor it happens. This strategy allows mentainance to be performed more efficiently, since more up-to date data is obtained about how closed th product is to failure.

2. Preventive Mentainance

This is the type of Mentainance that is performed with the intend of avoiding failur ~~Safty~~ Safety Violations Unnecessary producti Cost and losses and to Conserve Original materials of fabrication. The effectivenes of a preventive mentainane schedule depends on the analysis which it was base on the ground rules for cost efficiency

3. Corrective. Mentainance

Corrective Mentainance is a type of Mentainance Used for equiptment after equiptment ~~breakdown~~ breakdown or malfunction. Hence is often most expensive not only that it can worn equiptment but can damage other parts and cause multiple damage.

4. Evolutionary Computer maintenance:

This Concept has to do with evolution of computing. The replacement of old ~~infrastructure~~ infrastructure, hardware and operating system by newer, more modern and Innovative system.

Harzards threathening Computer System:

1. Dust
2. Temperature of the Sarrounding Medium,
3. Lost/damage of file,
4. Virus,
5. Power fiactuation,
6. Power Surge,
7. Static Electricity.

Remedy to the Harzards: (i) Dust;

- When purchasing a Computer rememeber to also purchase dust cover.
- Remember cleaning the Computer in regular basis.
- Place your Computer in an area that ~~it~~ it will not accomodate the easy transmition of dust to the computer.

- Placing the Computer near Window is not advisable, as this will make the Computer most Vulnerable to dust air/rai

2. **Temperature of the Sarrounding Medium:**

Place your Computer in an Area that is Cool. An air-Conditioned room is preferable.

3. **Lost/damage of File:**

If you regularly make backup Copies of your files and Keep them in a Separate Place, you Can get Some, if not all of your Information back in the event Somthing happens to the Originals of your Computer. Anything you Can not get easily should be at the top of your list. Make a check list of files to back up. It will help you determine what to back up and also give you a reference list in the event you need to retrive a backed-up file, for example

* Bank records and financial Information
* Digital Photographs
* Software you purchased and downloaded from Internet.
* Personal projects
* Your e-mail address book
* Your Microsoft Outlook Calendar
* Your Internet Explorer bookmarks

e.t.c.

Electrostatic Discharge (ESD):

Electrostatic discharge is caused by the buildup of electrical charge on one surface when they come in contact. This discharge comes with high voltage but low current and it may damage the following;

- Ic
- CPU
- Memory
* Cache chips
- expansion cards

Ways to deal with ESD:

1. Reduce its buid up by increasing the relative humidity of the room where the computer is located.
2. Touch the metal of your case when it is plugged in. This will drain off any static electricity buildup.
3. Avoid wearing socks on a carpeted floor.

Power Flactuation:

This is the variation of power supply below or above normal value. This will also affect the normal operation of Pc. In fact it may lead to damage of some vital components of the Pc.

Remedy

1. A stable power supply should be obtained by connecting a stabilizer from the mains to the computer.

2. Uninterupted power supply module (UPS) should also be connected.

3. Make sure that the element of your power Supply Unit e.g ~~Fan~~, heat Sink.. are also working so that the effect of rise in temperature will not also damage some other components withing the power parl ~~eliminated~~.

Power Surge:

This is temporary Increase of Voltage that can just last for few thousand Second's but in this time, the voltage can increase above the norminal Voltage e.g from 220V to 1000V or even higher.

Remedy to power Surge

Get a good power surge detector/protector to protech your Pc.

Virus

A Computer Virus is a type of malicious Software, or malware, that spreads between Computers and cause damage to data and Software. Computer Viruses aim to disrupt System, cause major operational issues, and result in data loss and leakage.

Virus can be categorized based on where they are located/they Hide themselves. The most common Virus type is the file virus; File virus hide themselves in executable files, when the executable file is run, the Virus is activated. Another type is macro Virus. These viruses attached themselves to the portions of application and disguise as macros. macros is simply an automated process within an application such as reading and automatically updating a date file or searching for and formatting specified text. Another type of virus is a boot sector. Virus. This type of virus hides itself in the MBR and is activated during startup when the MBR is located and initiated. ~~init.~~

Remedy:

1. Variety of antivirus utilities are available from third parties, such as Norton and McAffee Windows 2000 includes a native antivirus utility called AVBoot.
2. If an antivirus utility has fail to detect and remove a virus and you suspect the virus is limitted to the boot sector, use the FDISK/MBR command. This command will replace the infected MBR with a good copy from a floppy disk

REVIEW QUESTIONS

- Define and ~~state~~ state any five Importance of maintenance.
- List and briefly describe the four types of maintenance.
- Give two Remedy to the each of the following;
  - i. Power flactuation
  - ii. Dust
  - iii. ESD
  - iv. Virus destruction
- List any five modules that can be affected by the effect of ESD.

PC UPGRADE

An upgrade is a term that describes adding a new hardware or software in a computer that improves its performance. For example, with a hardware upgrade, you could replace your hard drive with an SSD or upgrade the RAM, providing a boost in performance and efficiency.

Basic Parts of a Computer:

- The Computer case or System Unit
- Motherboard
- Central Processing Unit
- Random Access Memory (RAM)
- Graphics Card or Graphics Processing Unit
- Sound Card
- Hard Disk Drive (HDD)
- Solid State Drive (SSD)
- Power Supply Unit (PSU)
- Monitor or Visual display Unit
- Keyboard.

Advantages of Pc Upgrade:

1. Performance Increase, which makes the overal Computer run faster and more Smoothly.
2. Capacity Increase. For example, adding a large hard drive ~~allo~~ allows the Computer to store more Information. Adding more memory Increase the Computers ability to run more programs efficiently.
3. You Can meet a program or games system requirement

Disadvantage of Pc upgrade.

1. Damage during Installation
2. When upgrading major Computer Components, such as a hard drive or Mother board you may nee to reinstall all your Software, a significat tim Investment.

__NB__ Upgrading the computer all depends on the type of Computer you have and what you ho to achieve with the upgrade. For example, th most Common upgrade to a Computer is upgrading the Computer memory to achieve better performance and greater memory capacity.

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Things to Consider when upgrading Software

1. **Compatibility:** there a many things to Consider when dealing with Compatibility. For One, your Software may rely on hardware that isn't Compatible with the new update.

2. **Community:** Joining a Comunity help you Invest In Others and allow you to learn new things and find Solution to your problem. It even allow us to learn new things and find answers to Some Some Solutions.

3. **Support:** when upgrading your Software make Sure to take in to Consideration where you can find Support.

Review Questions

- List any Ten (10) basic parts of a Computer.
- State three (3) advantages and two (2) disadvantages of Pc ~~upgrade~~ upgrade.
- list the three (3) basic things to Consider before making an upgrade.

Computer Power Supply Unit (PSU).

If there is any one component that is absolutely vital to the operation of a computer it is the power Supply Unit without it, a Computer is just an inert box full of plastic and Metal. The Power Supply Unit, also known as a PSU, converts the alternating Current (AC) line from your home to the direct Current (DC) needed by the personal Computer. In this article, we will learn how PC power supplies work and what the wattage rating mean.

In a personal Computer (PC), the Power Supply is the metal box usually found in a Corner of the Case. The Power Supply is visible from the back of many systems because it Contains the Power Cord ~~receptacle~~ receptacle and the coolling fan.

A typical PSU will have intergrated Connector to Send Power to the motherboard, microprocessor and SATA Storage.

Laptops and mini-PCs usually have their Power Supplies separate from the Computer assembly, Instead of integrated in their charging Cables. Power supplies, often reffered to as "Switching Power Supplies", use Switcher technology to Convert AC input to lower DC voltages.

The typical voltages supplied by PSU are;

* 3.3 Volts
* 5 Volts
* 12 Volts

- The 3.3 and 5 Volts are typically used by digital circuits like RAM, ROM, CPU etc.
- The 12 Volts is used to run motors in the disk drives and fans.

Power Supply Problems

- A typical failure of a PC power supply is often noticed as a burning smell just before the computer shuts down.
- Another problem could be the failure of the vital cooling fan, which allows components in the power supply to overheat.

Failure Symptoms of PSU

failure symtoms includes random rebooting or failure in windows for no apparent reson.

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REVIEW QUESTIONS

- What are the three (3) typical Voltages supplied ~~B~~ by PCs PSU

- The medium and the minimum voltages supplied by PSU are used to run which modules

- The maximum voltage supplied by PSU is used to run what

- Briefly describe any two (2) power supplies Problem.

- State the failure symtom of Power supply Unit (PSU)